

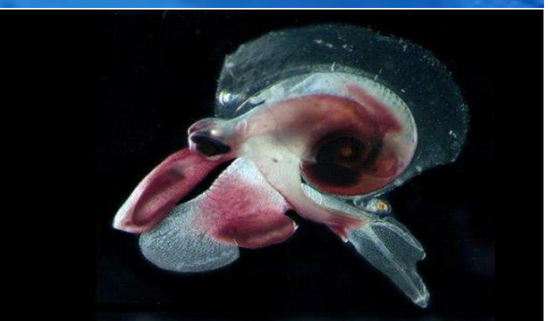

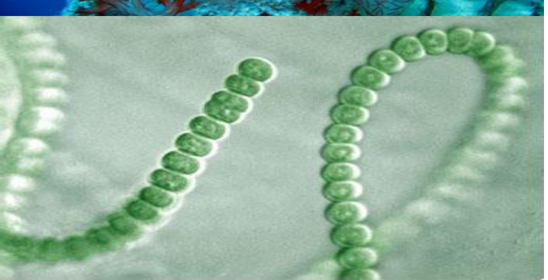

Marine Biology
海洋生物学

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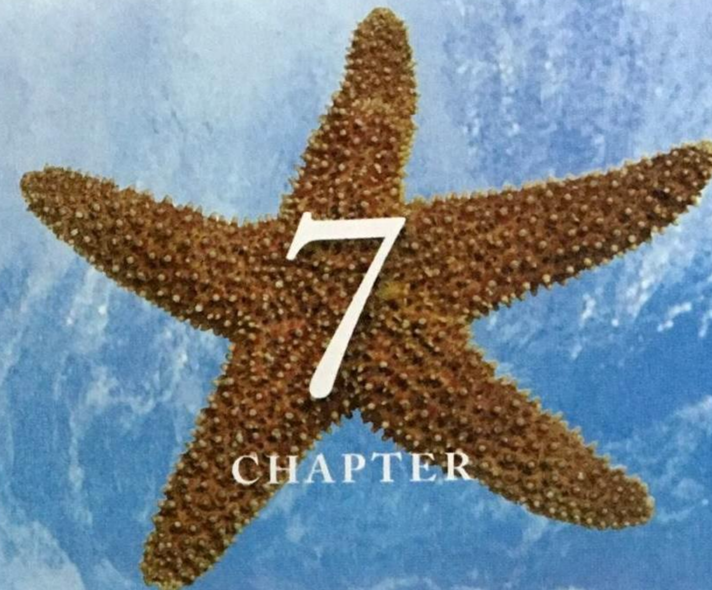
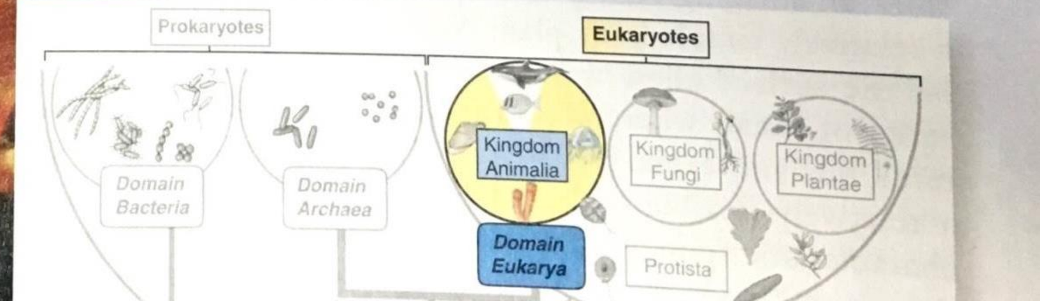


CHAPTER 7

Marine Animals Without a Backbone



- Major Phyla of Marine Invertebrates

1. Sponges (**海绵动物**)
2. Cnidarians (**腔肠动物**)
3. Flatworms (**扁形虫**)
4. Ribbon worms (**纽虫**)
5. Nematodes (**线虫**)
6. Annelids (**环节动物**)
7. Sipunculans (**星虫动物**)
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16. Hemichordates (**半索海生动物**)
17. Tunicates (**被囊动物**)
18. Lancelets (**文昌鱼**)

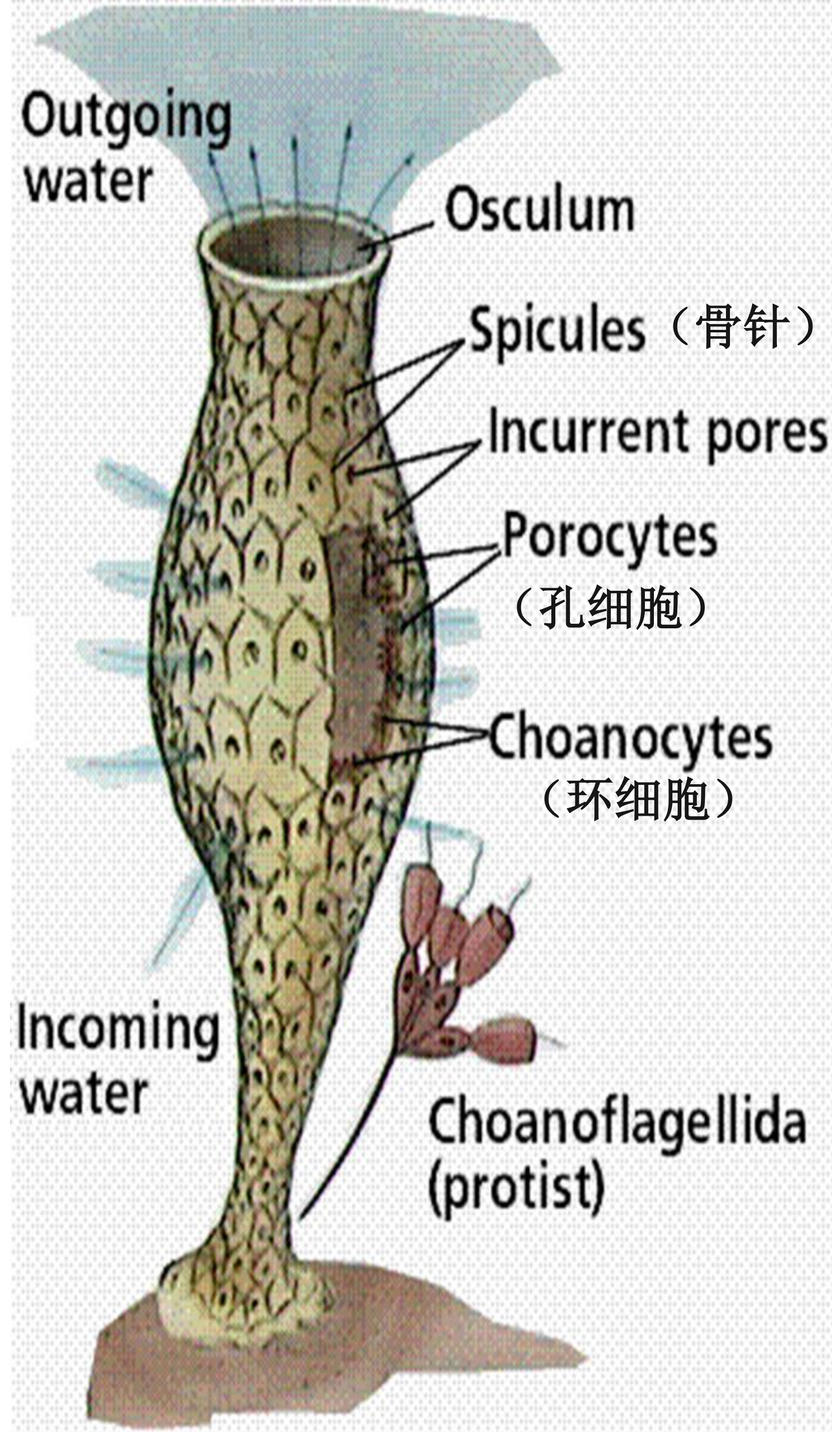
2. The Sponges

2.1 Characteristics of Phylum Porifera

- Sponges are **benthic creatures**, found at all latitudes beneath the world's oceans, and from the intertidal to the deep-sea
- Vary in size, shape and color, which are determined by the bottom sediment, the material they are growing in, algae they captured and ocean currents. Red, yellow, green, orange and purple are extremely common
- All adults sponges are **sessile** (non-motile and living attached) and some encrust on hard substrate, living solitary or colonial
- Some sponges bore into the shells of bivalves (mussels and clams), gastropods, and the colonial skeletons of corals by slowly etching away chips of calcareous material

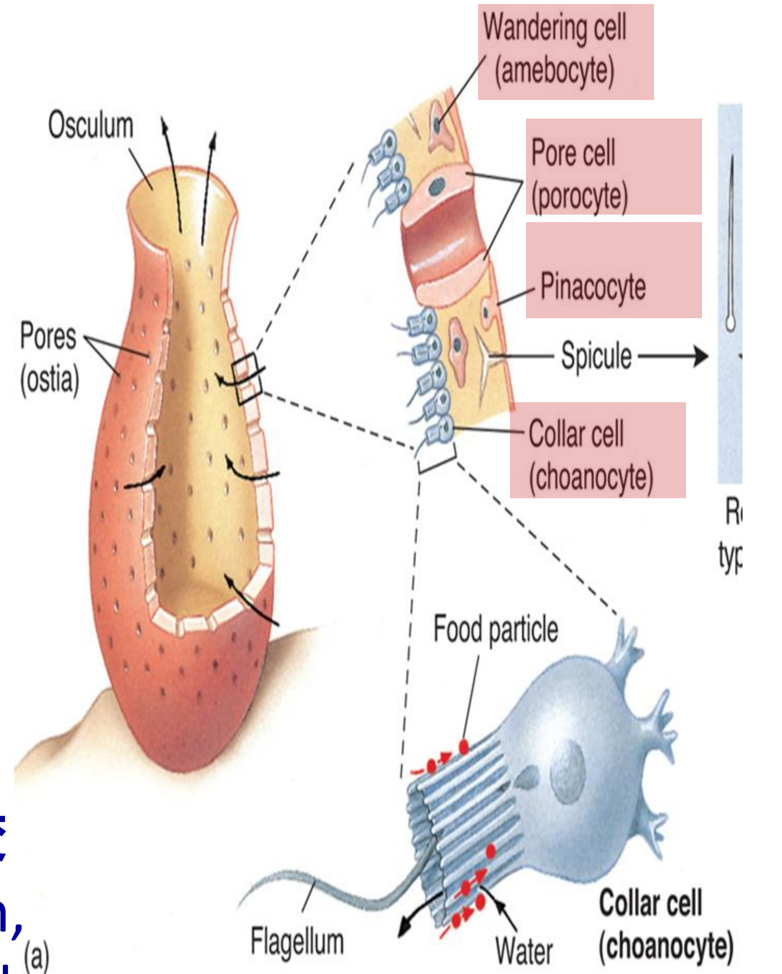
– **Asymmetric body with no true tissues or organs**, just aggregations of specialized cells built around a system of water canals

– Body full of tiny pores called **incurrent pores** or **ostia (进水小孔)** in which water passes through and enters large cavity called **spongocoel (海绵腔)**, finally exits out the **osculum (排水孔)**



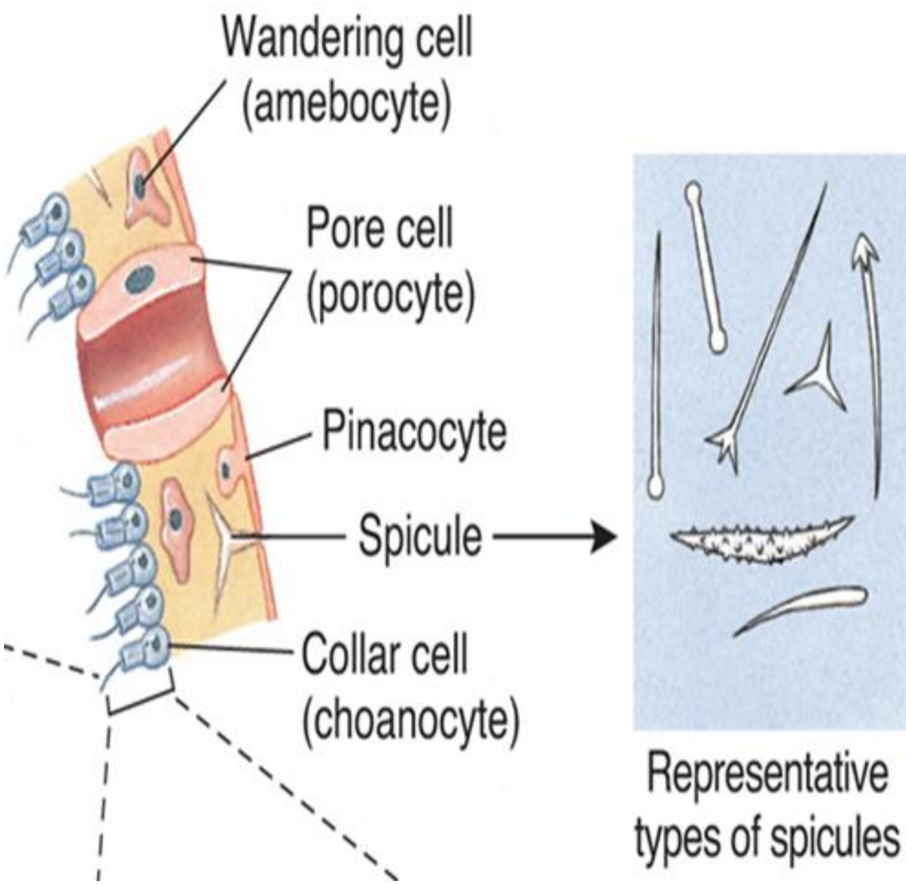
2.2 Types of Cells:

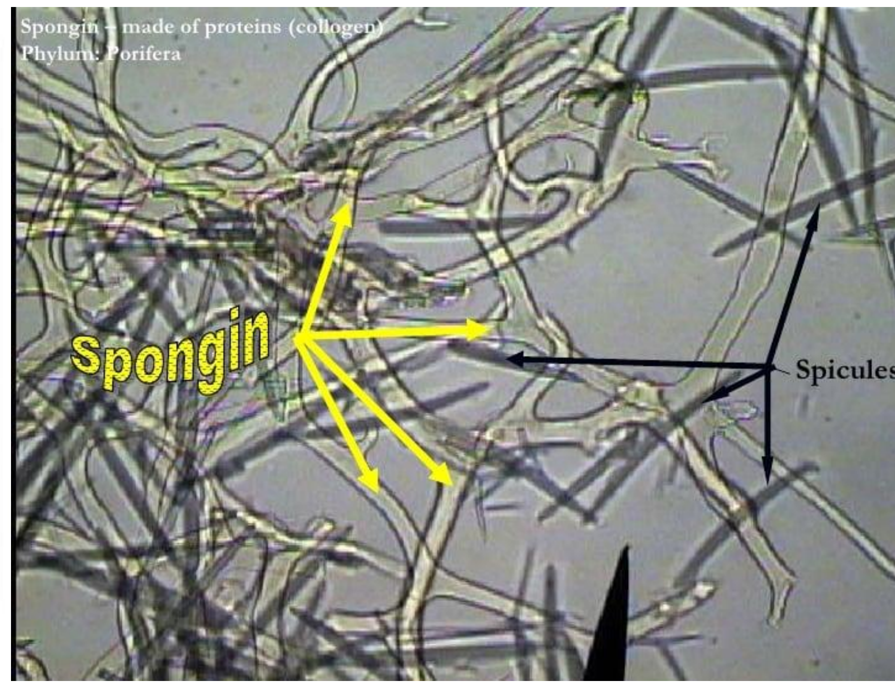
- **Choanocytes or collar cells (环细胞) :**
  —line interior canals of the body;
  —flagella create a water current that brings more food particles into the body , “Collars” traps food particles
- **Pinacocytes (扁平细胞):** flattened cells cover exterior of body
- **Porocytes (孔细胞):** cells with a pore to allow water to pass into body
- **Wandering cell (amebocyte, 游走细胞/变形细胞):** secrete the spicules and spongin, transport and store food particles, quickly repairing damages



2.3 Structural support

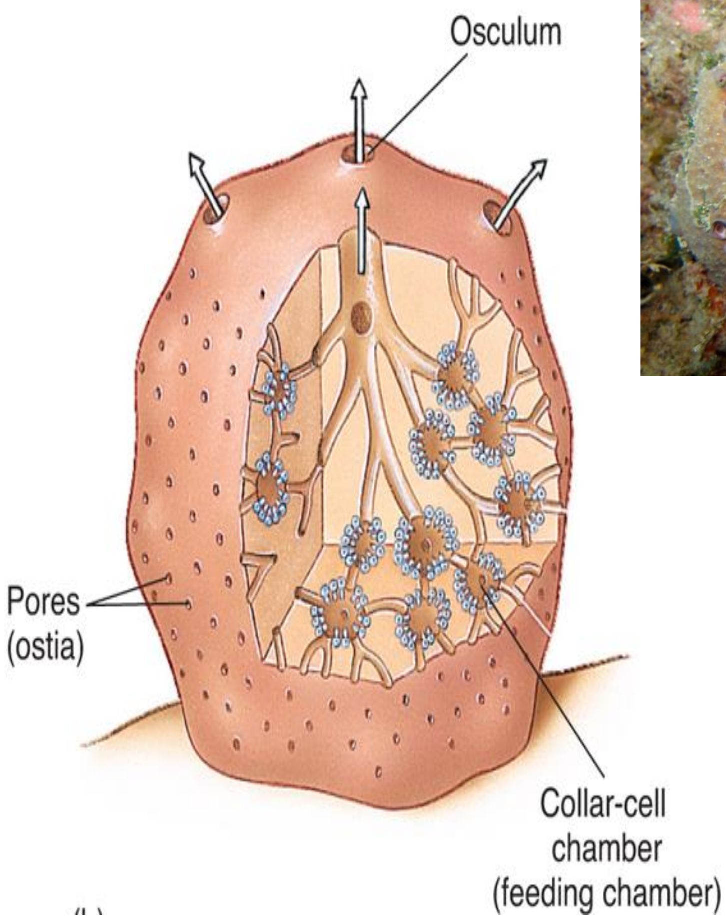
- **Spongin (海绵硬蛋白,** support protein) and **Spicules (骨针,** support structures made of silica or calcium carbonate)
- having a variety of shapes from simple rods to star-shaped

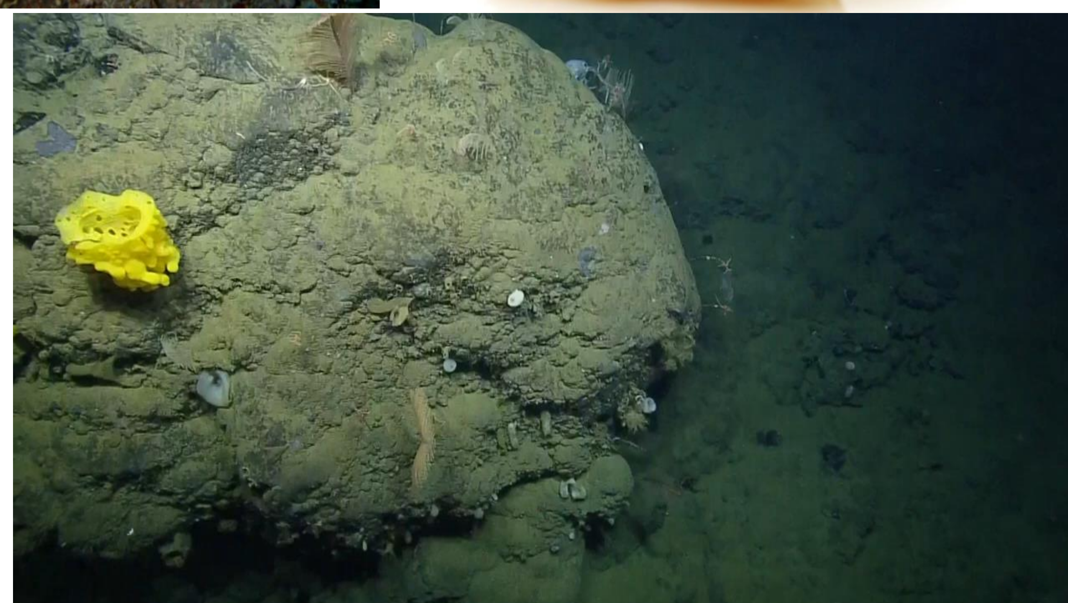




2.4 Complex structure

- Increased number of collar cells, complex canal system, increased size, which demands higher water flow through the sponge and therefore a larger surface area of collar cells

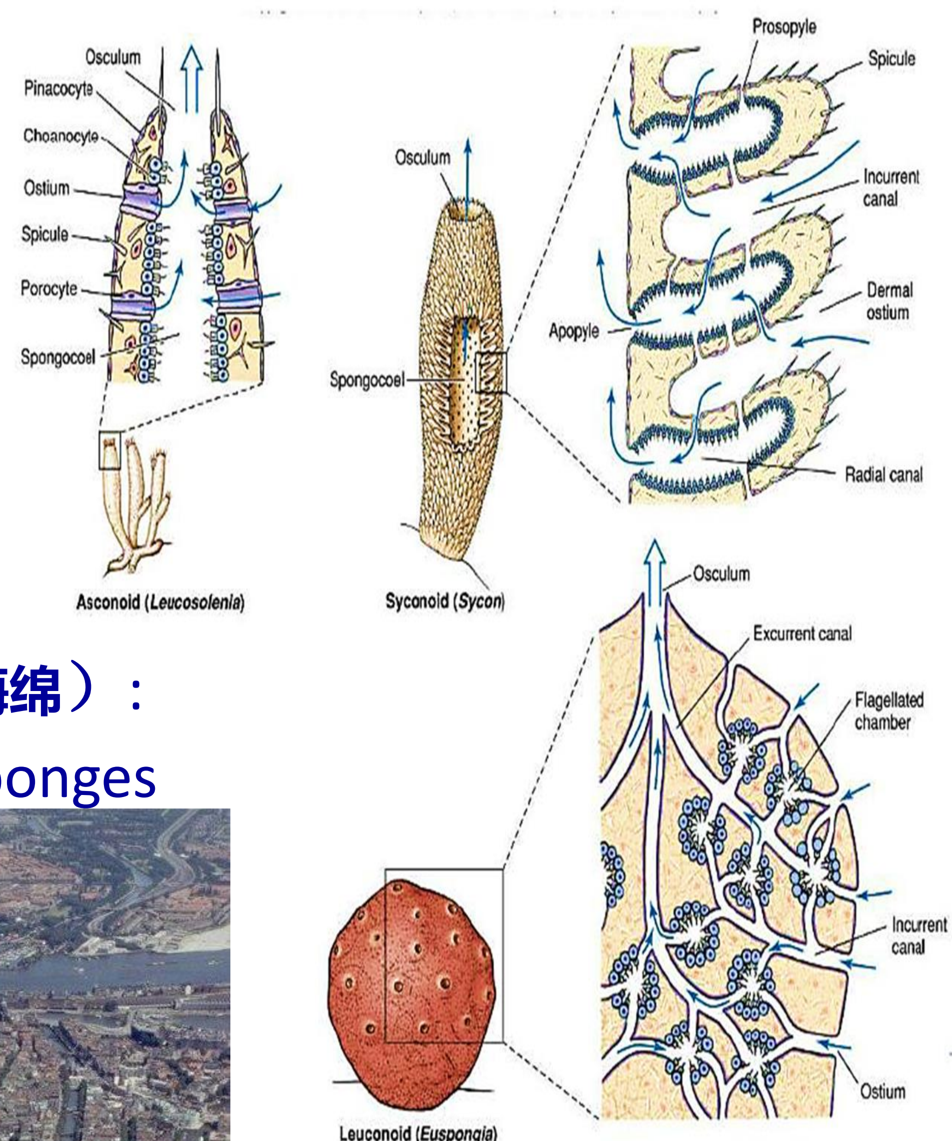




1) Asconoid sponge (**单沟系海绵**): the simplest form, tubular and small

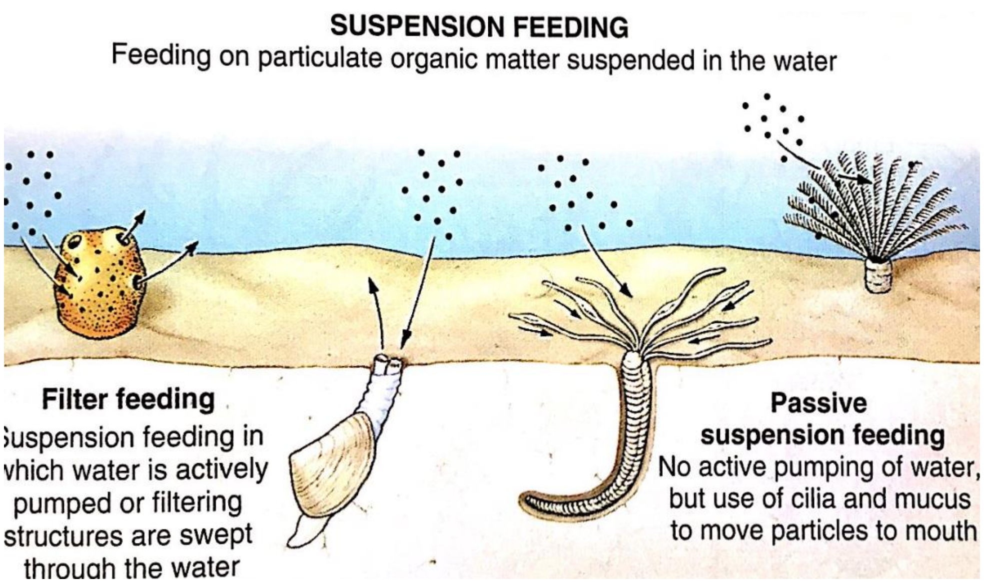
2) Syconoid sponges （**双沟系海绵**）: has body wall folding to it, but only a small amount

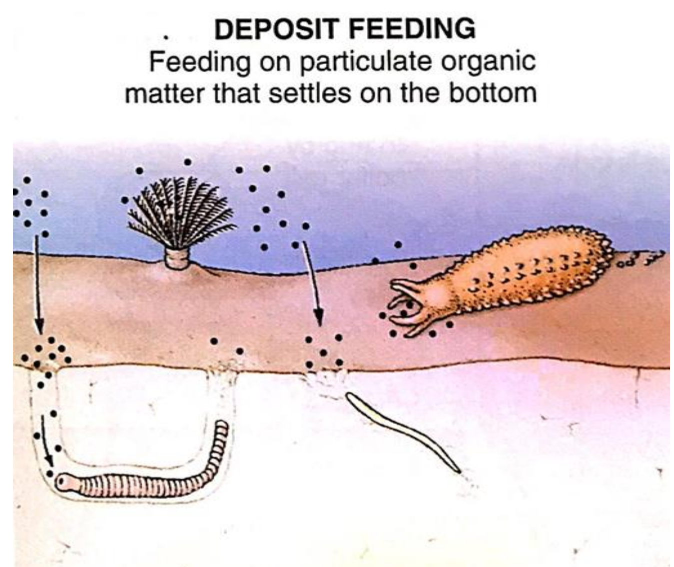
3) Leuconoid sponges（**复沟系海绵**）: many folds; largest type of sponges



2.5 Nutrition and Digestion

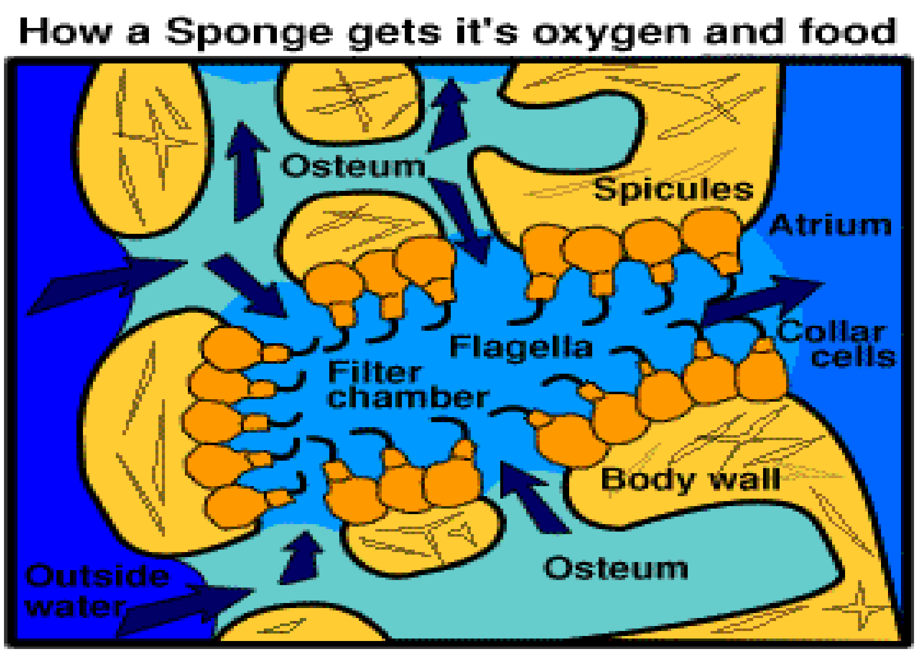
- **Suspension feeding（悬浮颗粒摄食）: feed on food suspended in water**
  - <u>Passive suspension feeder:</u> uses mucus or cilia to move food particles to the mouth
  - <u>Filter feeders:</u> filter their food from the water mainly by pumping the water into the internal cavity
- **Deposit feeding（沉积物摄食）: feed on organic matter that settles on the bottom**

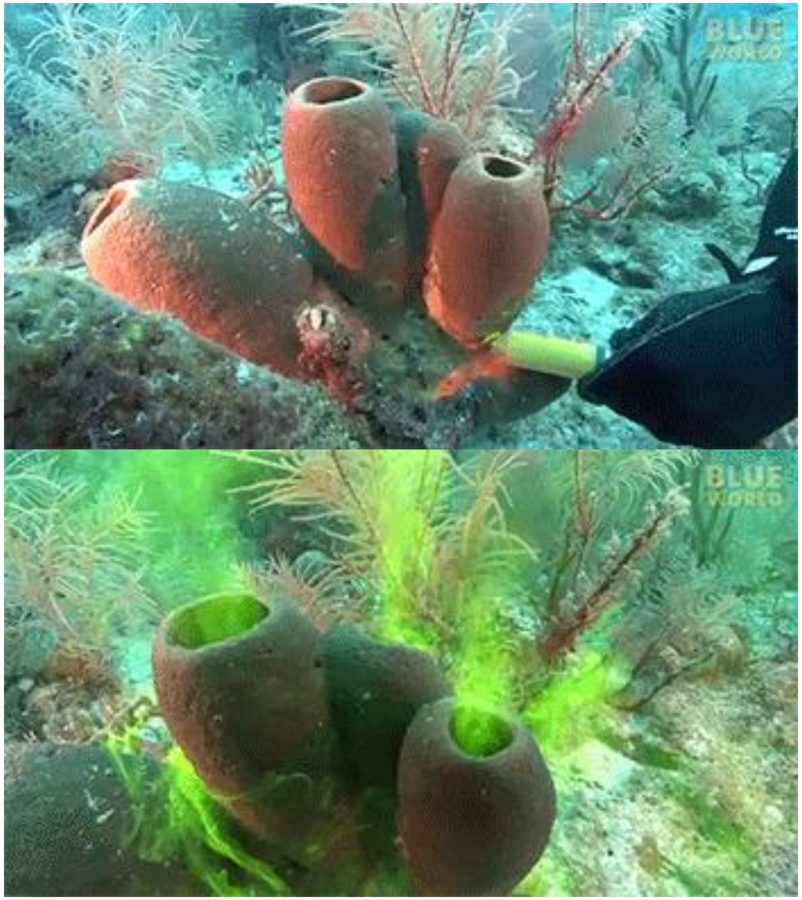




– The water movement is required to allow sponges to **filter feed** on plankton and dissolved organic matter in the water

– Water flow is also essential to carry metabolites (waste) away from cells and to carry gametes

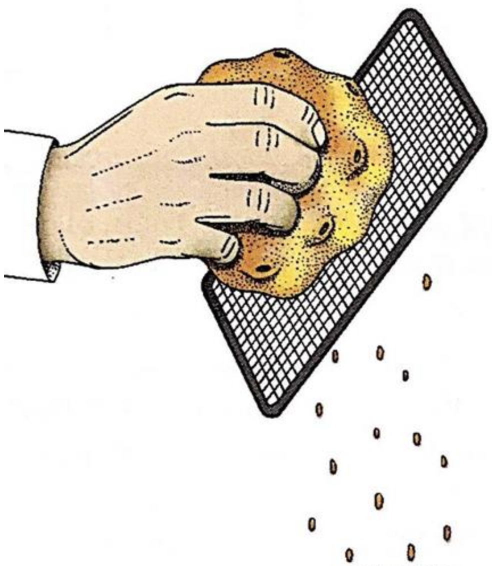


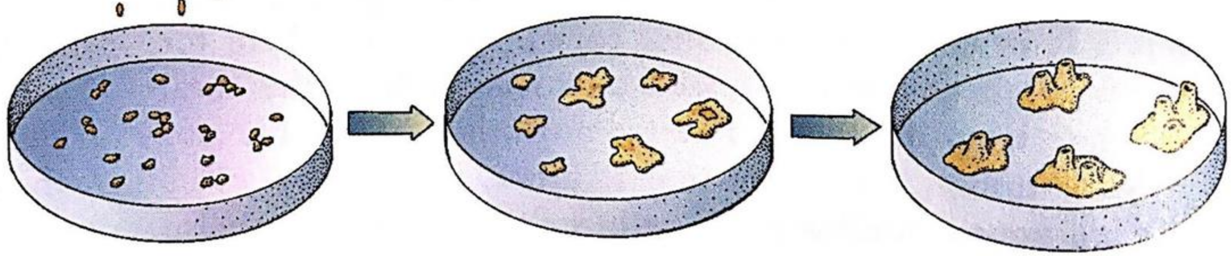


2.6 Modes of Reproduction

Depends on type of sponges

- **Asexual budding:** fragmentation of a cluster of cells from original sponge can begin growth in a new area
- **Regeneration**

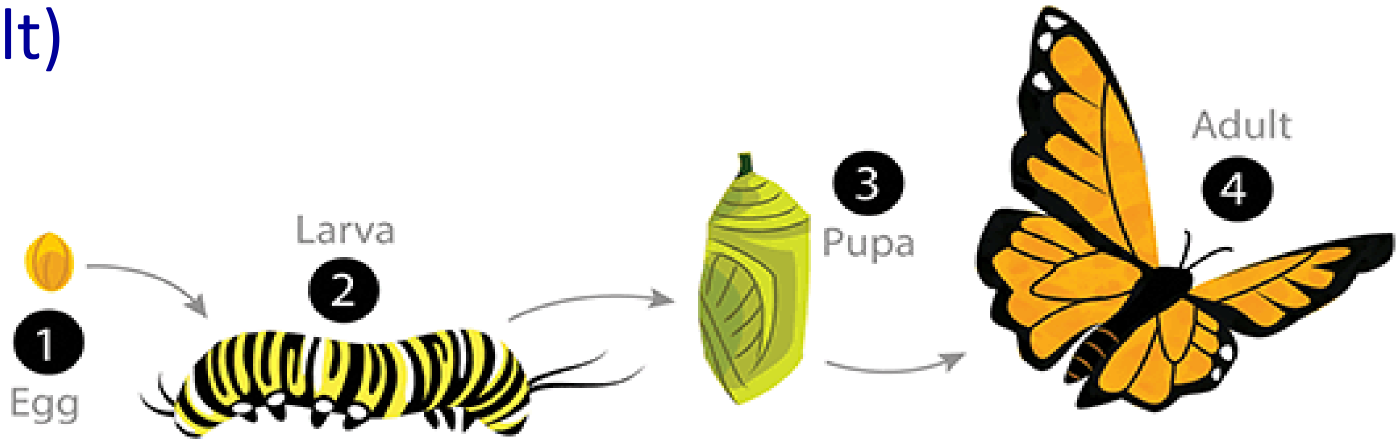


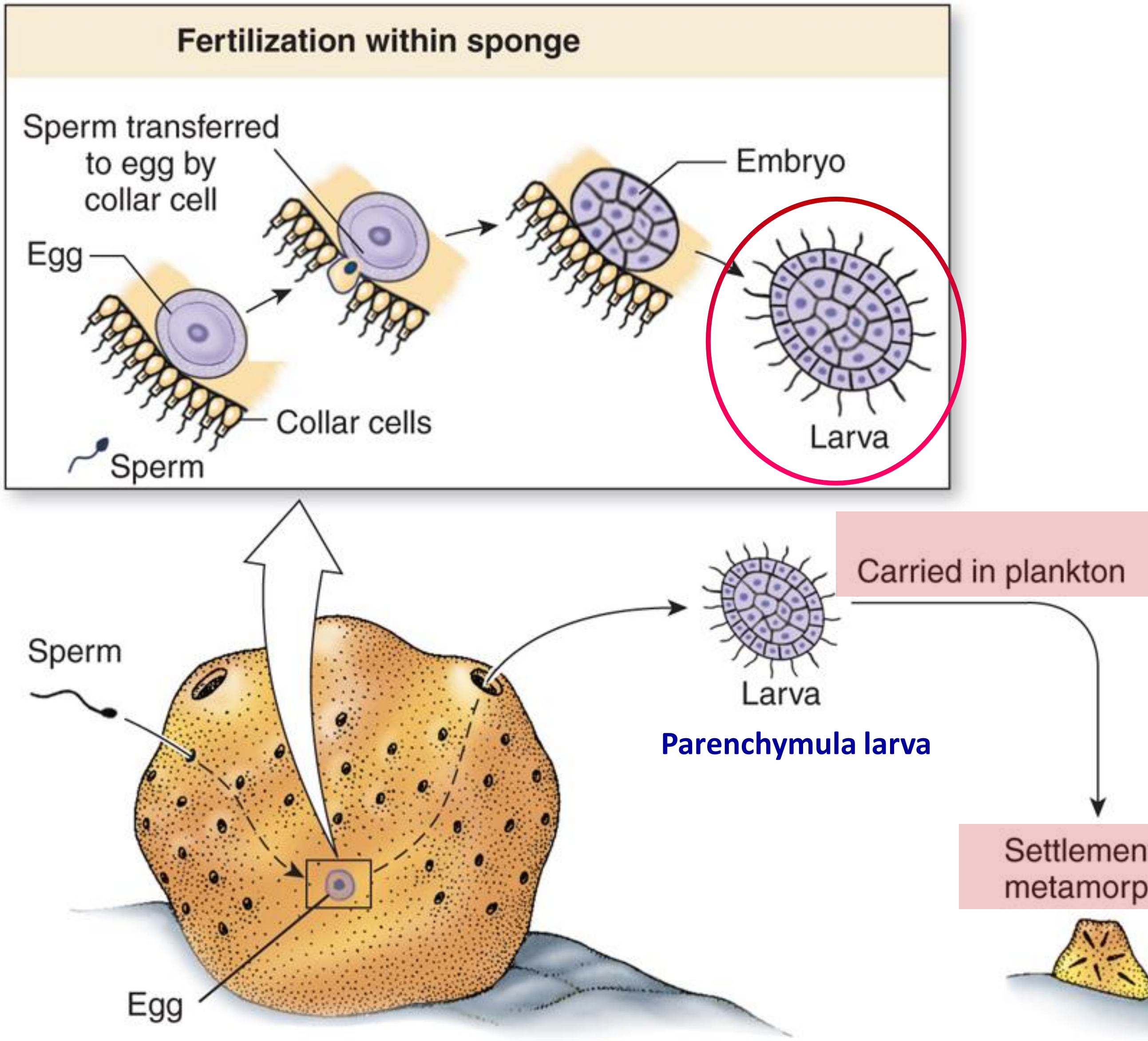


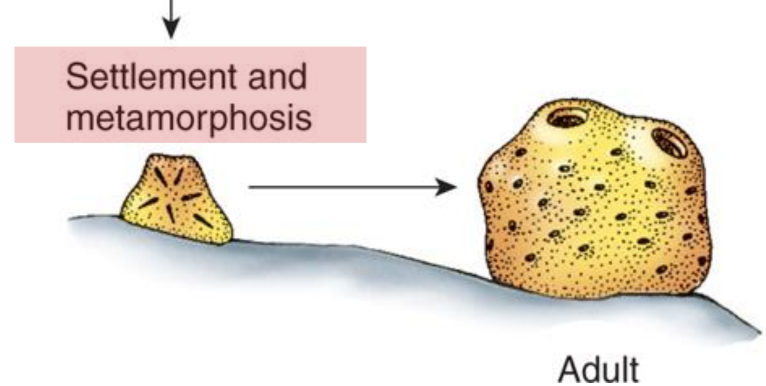
– **Sexual:** sperm are released into surrounding water (broadcasting spawners) to be picked up by a nearby sponge and directed to egg

– Most sponges are **hermaphroditic** (possess male and female reproductive parts)

– **Parenchymula larva (实胚幼虫)** is carried by currents until settled on the bottom to develop into a minute sponge (**Metamorphosis**, drastic change from larva to adult)




Fertilization within sponge
Sperm transferred to egg by collar cell
Egg
Collar cells
Sperm
Embryo
Larva
Carried in plankton
Larva
Parenchymula larva
Sperm
Egg
Settlement and metamorphosis
Adult



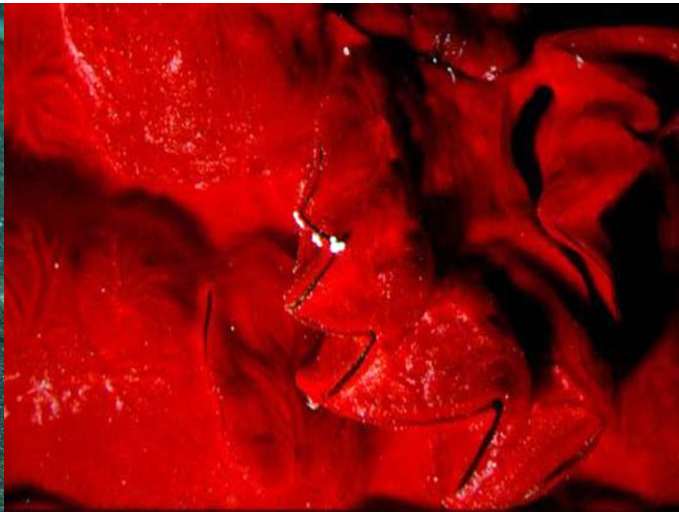
2.7 Ecological Role of Sponges

- Compete for space with other bottom dwelling organisms
- Need good material to attach to. If a fertilized baby sponge doesn’t get it, it dies
- Produce chemicals to kill other species around them to make room – good natural products resources
- Few animals, e.g. sea turtles, eat sponges- eating the spicules is like eating needles
  - Sea Turtle has a toughly lined digestive tract, its poop is composed of 95% of spicules

2.8 Sponges and Symbiotic Relationships

- Sponge houses bacteria, that provide food for sponge while the bacteria gains nutrients and a house， e.g. Cyanobacteria is a photosynthesizer, which produces its own food and shares with the sponge
- Sponges and other motile organisms: the sponge grows on the crab becoming motile. The crab gets camouflage





2.9 Sponge Classification

http://www.marinespecies.org/porifera/porifera.php?p=taxdetails&id=607239

Phylum — Class (纲) — Characteristics

Porifera

- Calcarea (钙质海绵)

  - **-$CaCO_3$ spicules**
  - - **all are marine**

- Hexactinellida (六放/玻璃海绵)

  - **-SiO_3 spicules**
  - - 6-rayed spicules
  - - **all are marine**
  - - "glass sponges"

- Demospongiae (寻常海绵)

  - **-SiO_3 spicules**
  - - may have spongin
  - - marine, brackish or fresh water

- Sclerospongiae (硬骨海绵)
  - - **coralline sponges**
  - - **all are marine**

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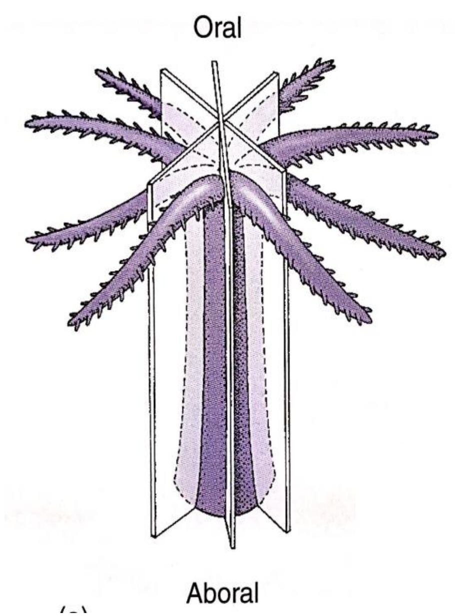
3. The Cnidarians (sting cells animals, Cnid: nettle)

3.1 Characteristics of Phylum:

- Radial symmetry, allows them to sting and capture organisms in any direction
- Mostly marine organisms, about 10,000 species known
- Generically, includes Corals, Sea Anemones and Jellyfish




Elk Horn Coral


Sea Anemone


Jellyfish

– Outer layer: **ectoderm/epiderm (表皮)** has cells that capture food and secrete mucus

– Jelly-like material between two layers: **Mesoglea (中胶层)**

– Inside layer: **endoderm (胃皮、内皮)** surround the central cavity

– Central cavity: **gastrovascular/internal cavity (消化循环腔)** which extend to **tentacles (触角)**